

From Static Rules to Reasoning Platforms:

Scaling Intelligent Canary Delivery in 2026



Daniel Oh

DEVELOPER ADVOCATE | JAVA CHAMPION | CNCF AMBASSADOR



@danieloh30



Developer Advocate at IBM (Red Hat)

- Cloud Native Runtimes
- AI, GitOps, Security, and Serverless



JAVA Champion



CNCF Ambassador & TAG DevEx co-chair



AAIF Ambassador



Microsoft MVP – Java & Developer Tools



Keynote Speaker & Published Author



Living in Boston, MA USA



Java[™]
Champions



INFRASTRUCTURE EXPLOSION

More complexity. More scale. More **ways to fail**.



KUBERNETES CLUSTERS



Kubernetes clusters have evolved into distributed **Edge-to-Cloud** meshes.

MICROSERVICE COUNTS (Since 2022)



2022

NOW

Microservice counts per organization have **tripled** since 2022.



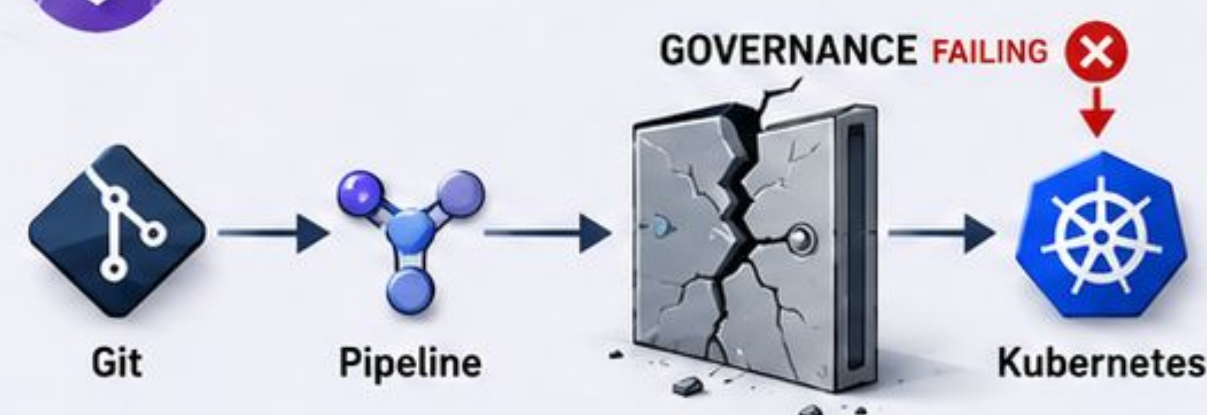
COGNITIVE LOAD & FAILURE



“Cognitive Load” is the **#1 reason** for deployment failures.



GITOPS & GOVERNANCE



Git

Pipeline

Kubernetes

GitOps is standard, but “**GitOps Governance**” is **failing**.



Scale without observability and governance is not growth — it’s a **risk multiplier**.

FRAGILITY OF THE “MAGIC NUMBER”



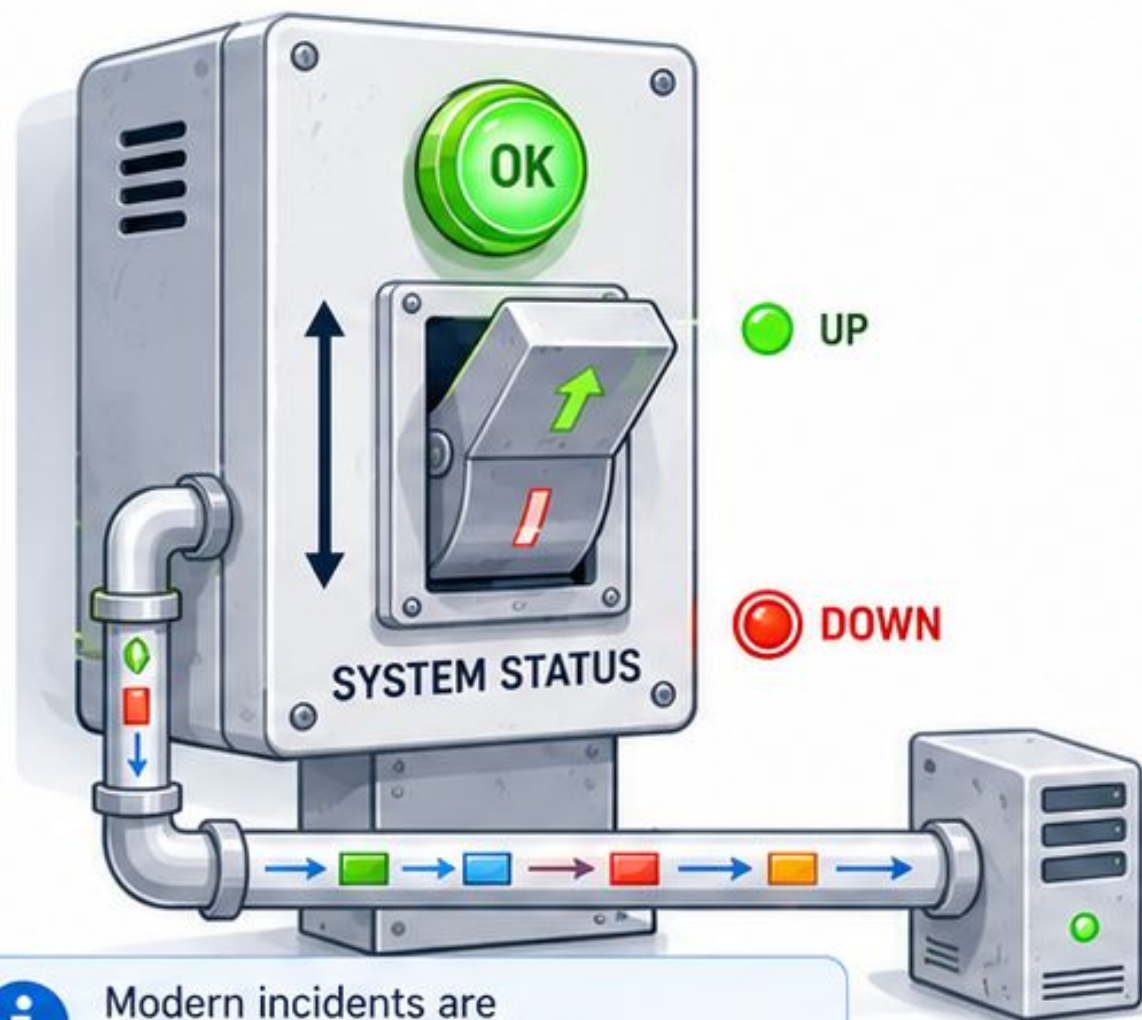
Perfect numbers can hide broken systems.
You need **depth**, **context**, and **corruption detection** — not just success rate.

2026 INCIDENT: GHOST IN THE MACHINE

In 2026, we deal with “Gray Failures.” The light is green, but the car is making a strange noise.

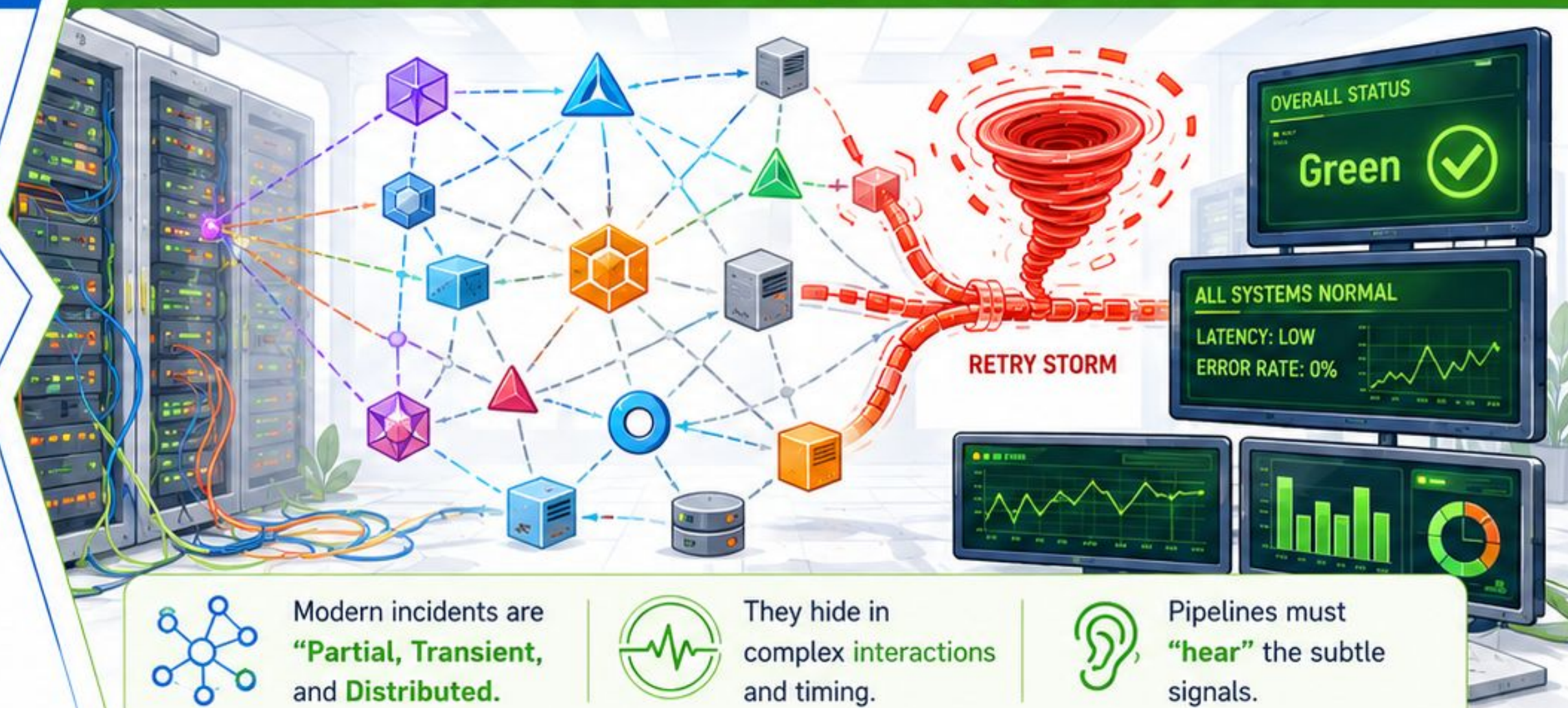
Our pipelines need to “hear” that noise.

THE OLD VIEW: “BINARY” (UP/DOWN)



Modern incidents are rarely “binary” (Up/Down).

THE NEW REALITY: “PARTIAL, TRANSIENT, & DISTRIBUTED”



Modern incidents are “Partial, Transient, and Distributed.”



They hide in complex interactions and timing.



Pipelines must “hear” the subtle signals.

CASE STUDY: THE “RETRY STORM”



Deployment passes unit tests.



Causes a “Retry Storm”.



Standard monitoring misses it (Shows “Green”).



DB crashes 2 hours later.

IMPACT



Standard monitoring misses it until the DB crashes 2 hours later.

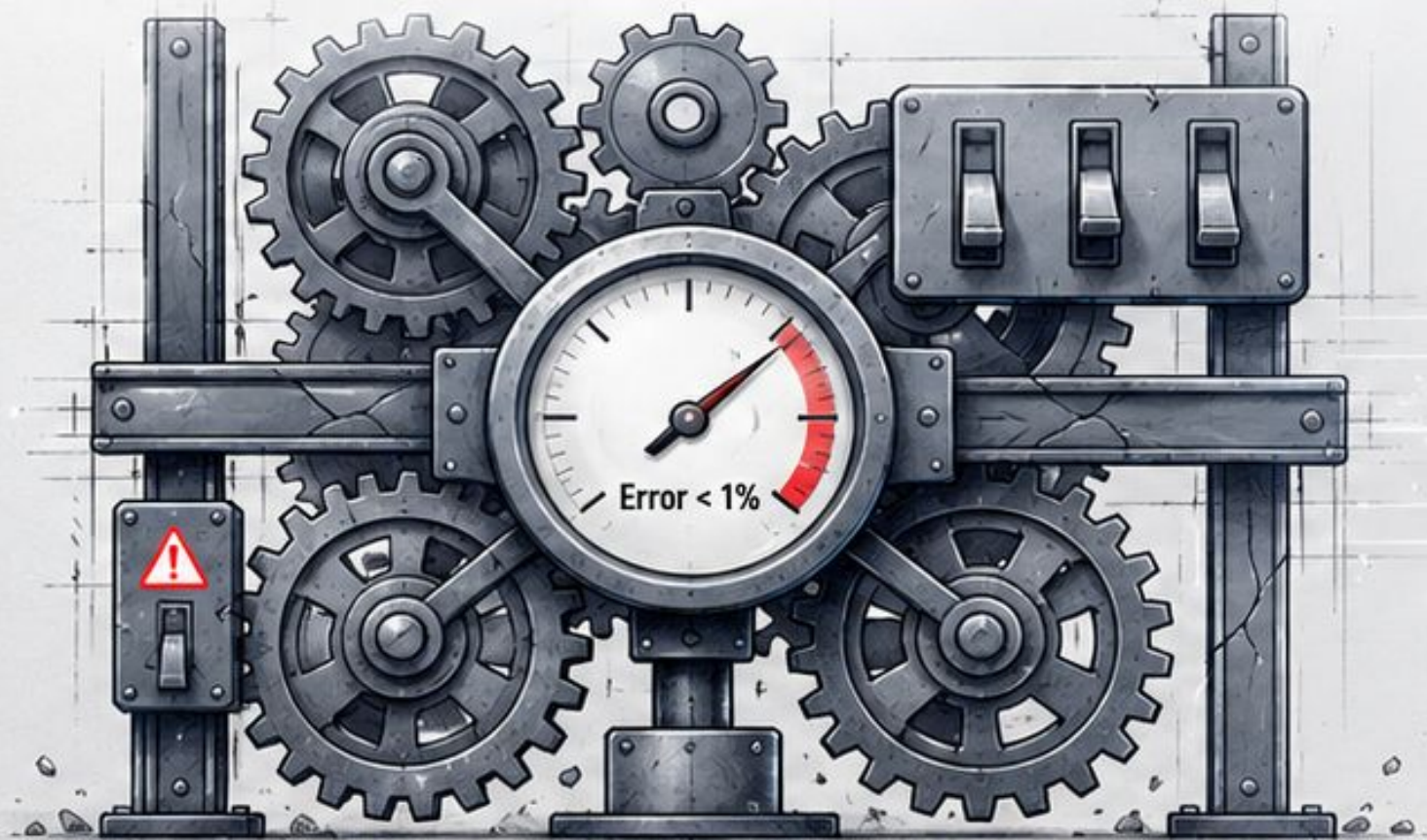
Impact Over Time



THE FUTURE OF RELIABILITY IS NOT JUST SEEING GREEN, IT’S HEARING THE NOISE.

REASONING PLATFORM IS

STATIC AUTOMATION



RULES • THRESHOLDS • FIXED LOGIC

Brittle. Reactive. Blind to context.

REASONING PLATFORM



CONTEXT • CORRELATION • REASONING

Intelligent. Proactive. Aware.



FROM STATIC TO REASONING

Shifting from brittle, fixed threshold queries (Static Automation) to systems that reason through data (Reasoning Platform).



THE "DIGITAL SRE"

AI agents that evaluate deployment health using human-like context, but process decisions in milliseconds.



ELIMINATING BOTTLENECKS

Eradicating "Alert Fatigue" and the need for manual promotion gates that slow down delivery pipelines.



TACKLING "GRAY FAILURES"

Autonomously detecting silent system degradation even when standard monitoring metrics appear completely "green".



From reacting to alerts, to understanding what's really happening. **That's the power of reasoning.**

TRANSITIONING TO AGENTIC DELIVERY

From static rules to intelligent reasoning. From manual rollouts to autonomous delivery.

STATIC VS. REASONING APPROACH

STATIC RULES



Static: "If error > 1%, Rollback."

REASONING PLATFORMS



Reasoning: "Errors increased, but they match a known upstream ISP outage. Continue rollout, but monitor latency."

PARADIGM SHIFT



From
"Metrics Thresholds"

FOCUS SHIFT

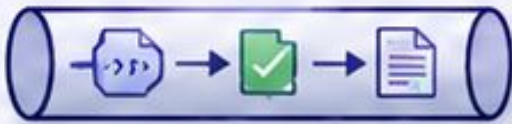


To
"Evidence Synthesis"

THE ROLE OF THE AGENT

CI

(Continuous Integration)



AGENT

CD

(Continuous Deployment)



Autonomous Logic Gate

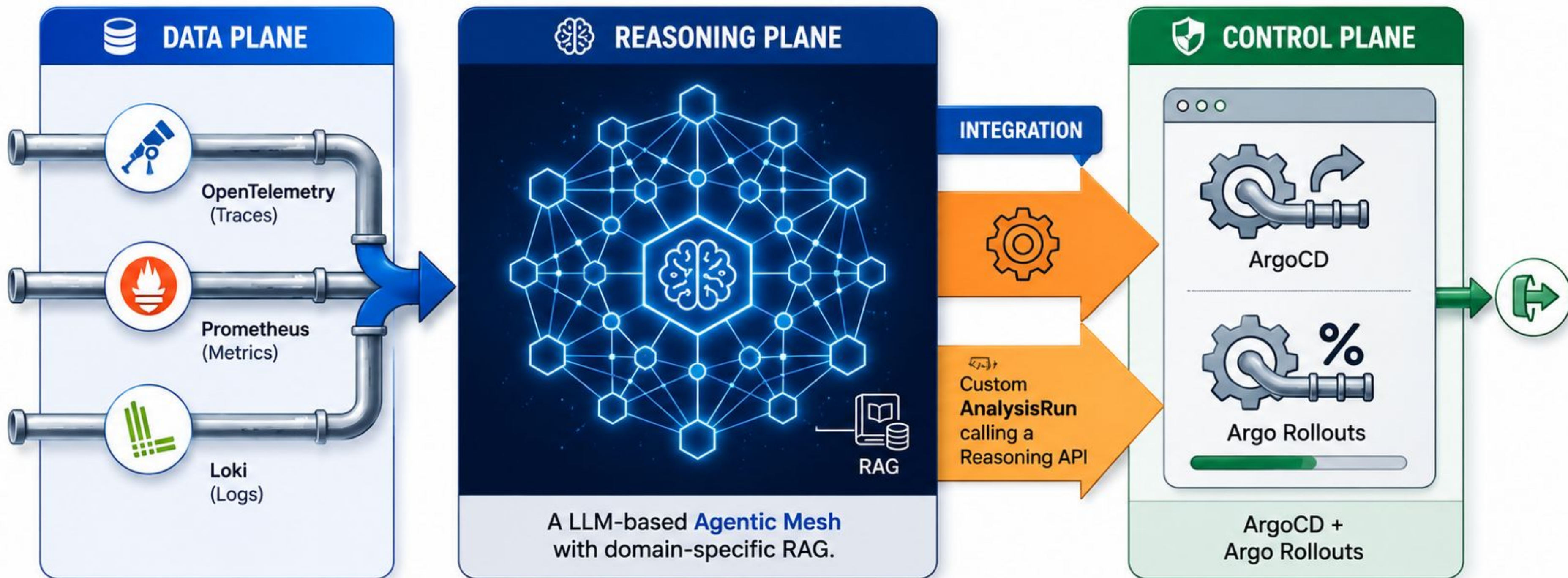
An autonomous logic gate between CI and CD.



Smarter decisions. Fewer false alarms. Continuous delivery, **intelligently.**

ARCHITECTURAL BLUEPRINT

Observability meets intelligence. Intelligence drives autonomous delivery.



UNIFIED OBSERVABILITY

Traces, metrics, and logs flow together into a single intelligent system.



INTELLIGENT REASONING

LLM-powered agents correlate signals and understand context with domain-specific RAG.



AUTONOMOUS DELIVERY

AI-driven insights trigger safe, automated actions via ArgoCD & Rollouts.



RELIABLE OUTCOMES

Continuous rollout with built-in safety, analysis, and feedback loops.

DIGITAL SRE PERSONALITY

An agent that defines, investigates, and decides.

1 PERSONA DEFINITION

- Defining “Persona” in deployment manifests.
- Example: “Aggressive” for Dev, “Conservative” for Production.



2 ACTIVE INVESTIGATION

- Ability to “Investigate”.
- The agent triggers extra profiling during a Canary.



3 HUMAN-IN-THE-LOOP

- The agent presents a “Case” for approval.



CONTEXT-AWARE

Personalized behavior based on environment and needs.



PROACTIVE INSIGHTS

Automatically investigates and surfaces hidden issues.



COLLABORATIVE DECISIONS

Human oversight ensures safe and confident outcomes.



FASTER & SAFER DELIVERY

Smarter rollouts with continuous learning and improvement.

Live Demo <https://github.com/danieloh30/progressive-delivery>

AI-Driven Progressive Delivery with Argo Rollouts



ARGO ROLLOUTS

Progressive delivery
made simple



KUBERNETES

Run and orchestrate
your workloads



QUARKUS

Agentic AI apps for
your Digital SRE



LANGCHAIN4J

Java library to integrate
LLMs and Vector Stores
into JVM apps



LANGCHAIN4J

Trigger rollouts, observe,
and iterate in real time



PROGRESSIVE DELIVERY

Canary, Blue/Green,
and more with
automated rollouts



AI INSIGHTS

LLM-powered insights
from logs, metrics, and
traces for smarter decisions



SAFE & RELIABLE

Automatically promote
or rollback based on
real-time signals



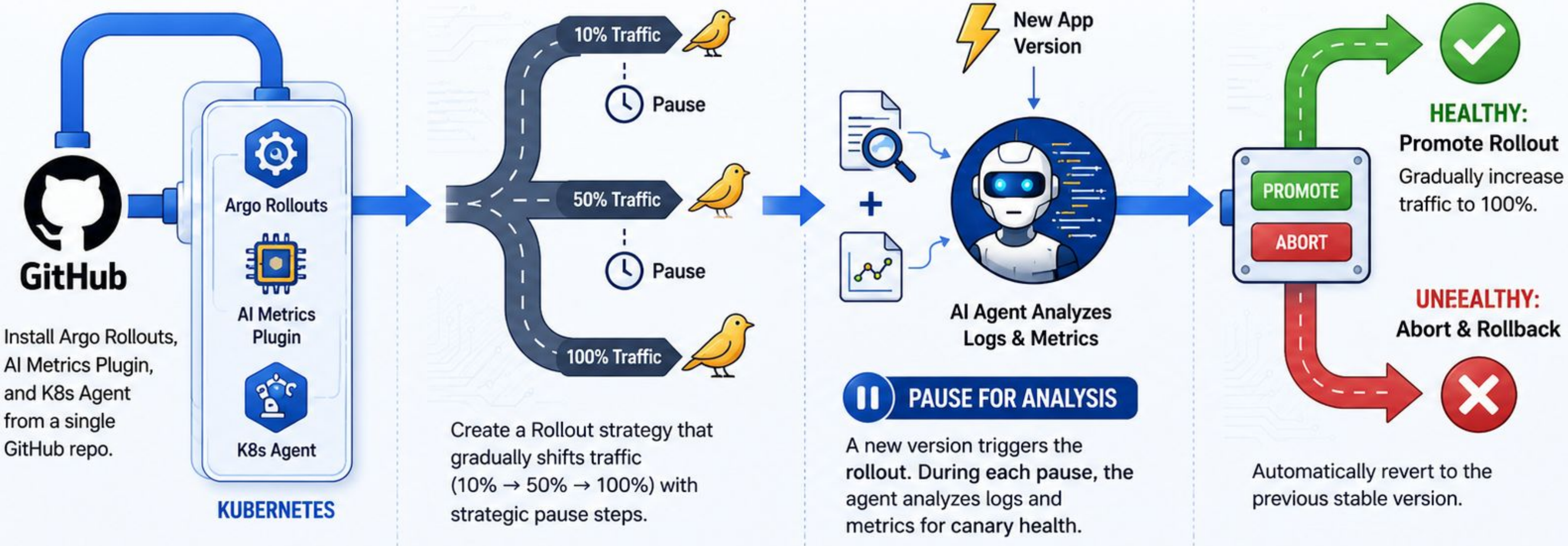
CONTINUOUS FEEDBACK

Observe, learn, and
improve with every
deployment



Ship Faster. Reduce Risk. Deliver with Confidence.

DEMO SCENARIO - INTELLIGENT CANARY DELIVERY



INTELLIGENT ROLLOUTS

Safe, gradual traffic shifts with strategic pauses.



AI-POWERED INSIGHTS

Agents analyze logs and metrics to detect issues early.



AUTOMATED DECISIONS

Promote healthy releases, abort risky ones.

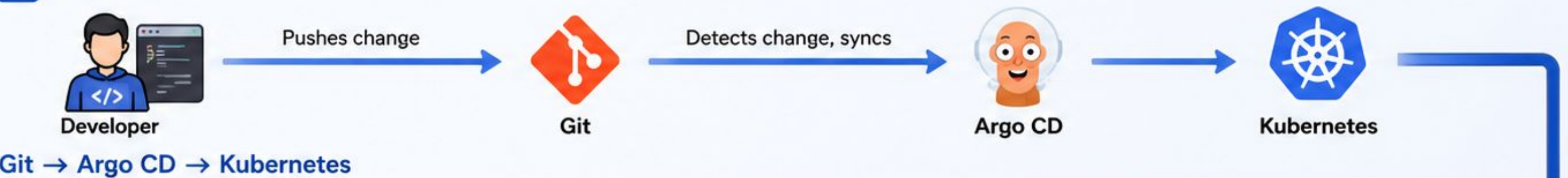


RELIABLE DELIVERY

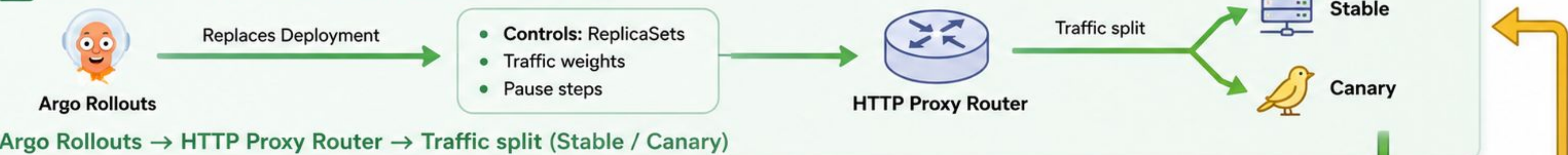
Higher velocity with confidence and rollback safety.

DEMO ARCHITECTURE - MULTI-LAYERED

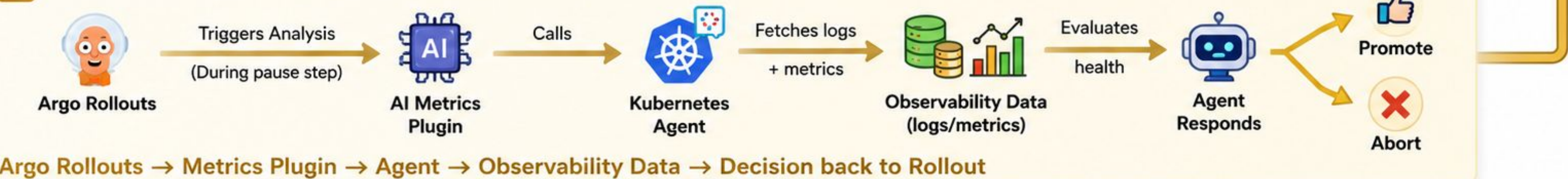
1 LAYER 1 — GITOPS CONTROL PLANE



2 LAYER 2 — PROGRESSIVE DELIVERY ENGINE



3 LAYER 3 — AI ANALYSIS & DECISION LAYER



GITOPS CONTROL
Declarative, versioned,
and auditable.



PROGRESSIVE DELIVERY
Safe rollouts with traffic
shaping and automated control.



AI-DRIVEN INSIGHTS
Real-time analysis of logs
and metrics for smart decisions.



AUTOMATED OUTCOMES
Promote healthy releases or
abort risky ones, automatically.

GUARDRAILS: KEEPING THE AI IN CHECK

Built-in controls that keep AI agents safe, reliable, and accountable.



KILL SWITCH



Disabling the Agent with one flag.



THRESHOLD CAPS



50% Error Rate
(Hard Fail)



AI can't override a "Hard Fail"
(e.g., 50% Error Rate).



SECURITY



Ensuring the Agent doesn't leak PII in its
"Reasoning Report."



RESOURCE QUOTAS



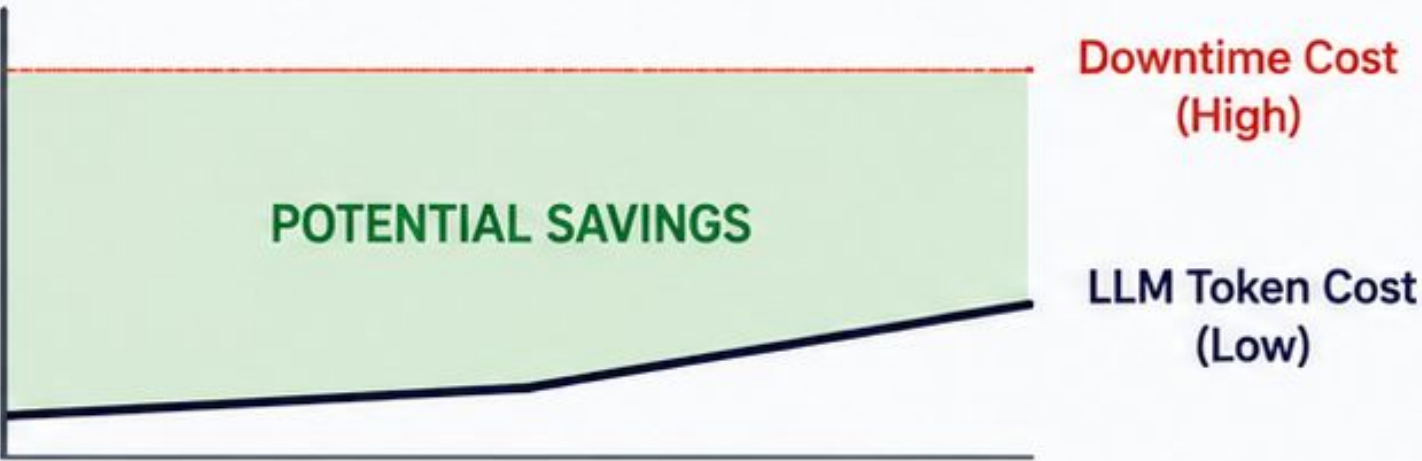
Preventing the Agent from
"Over-Searching" logs.

COST OF INTELLIGENCE

Smarter reasoning. Lower cost. Higher ROI.



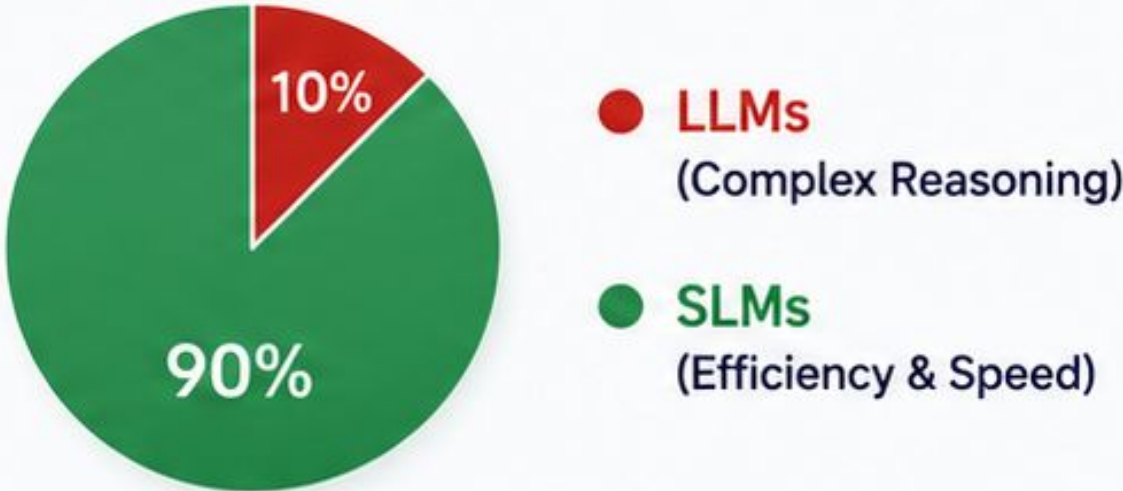
LLM TOKEN COST vs. DOWNTIME COST



LLM Token Cost vs. Downtime Cost.



OPTIMIZATION: SLMs for 90% of TASKS



Using Small Language Models (SLMs) for 90% of tasks.



EDGE REASONING: DIGITAL SRE IN CLUSTER



Running the 'Digital SRE' inside the cluster.



ROI CALCULATION: HUMAN CAPITAL



Reduced 'Burnout' and 'Developer Toil'.



LOWER COST

Reduce token spend and downtime impact.



HIGHER EFFICIENCY

SLMs handle 90% of tasks with speed and precision.



EDGE INTELLIGENCE

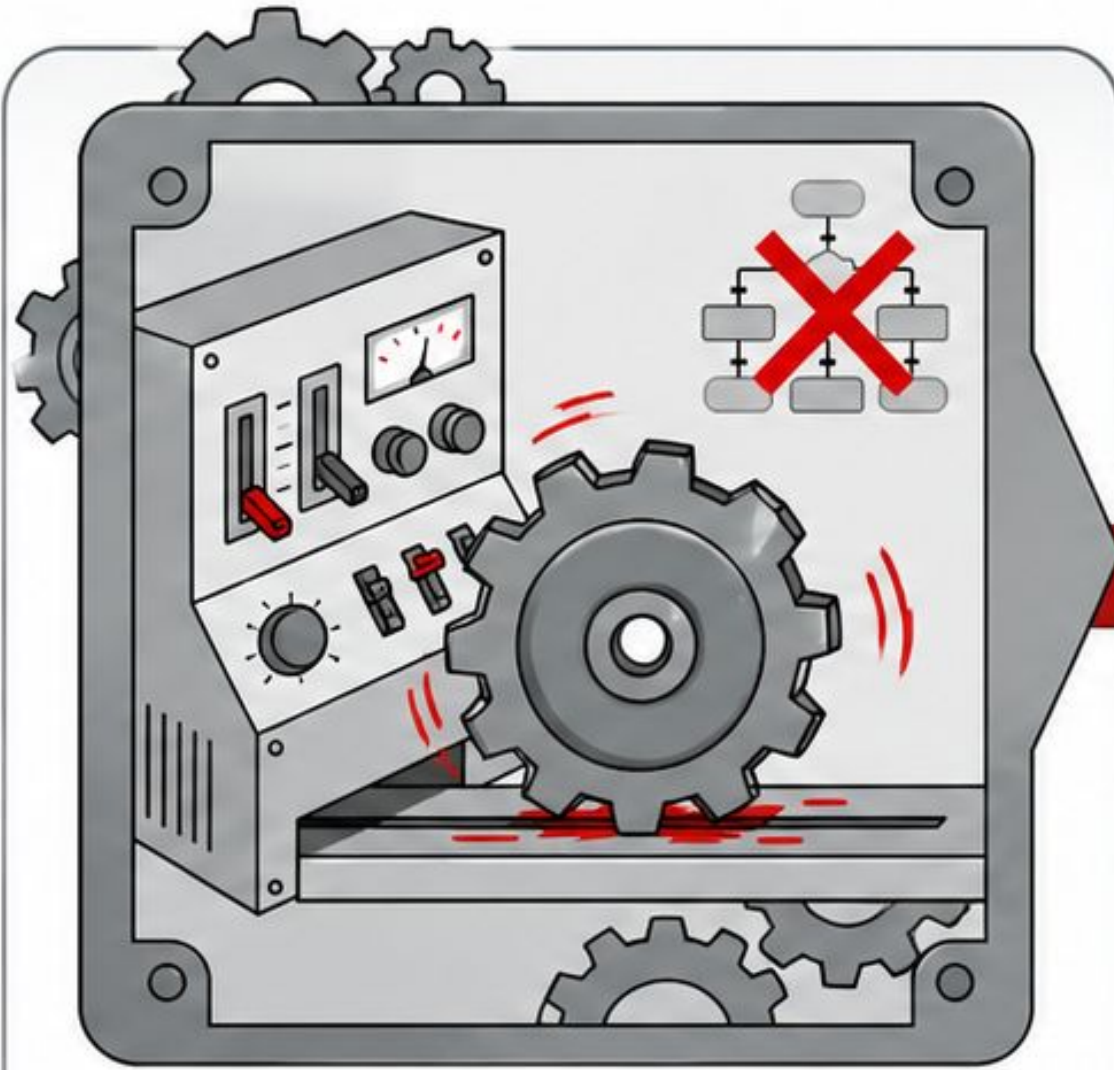
Digital SRE reasoning closer to the infrastructure.



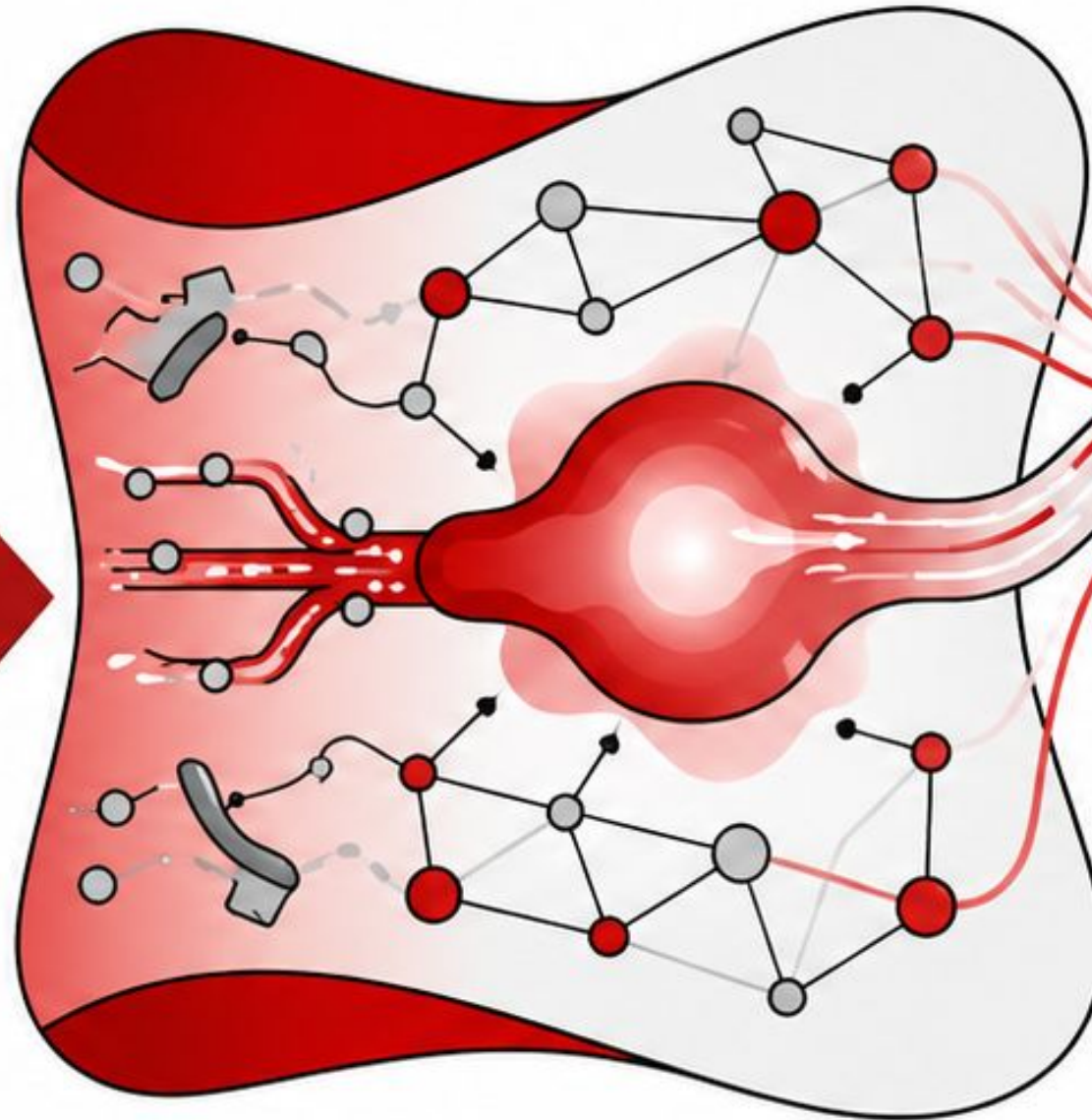
BETTER ROI

Significantly reduce human toil and accelerate delivery.

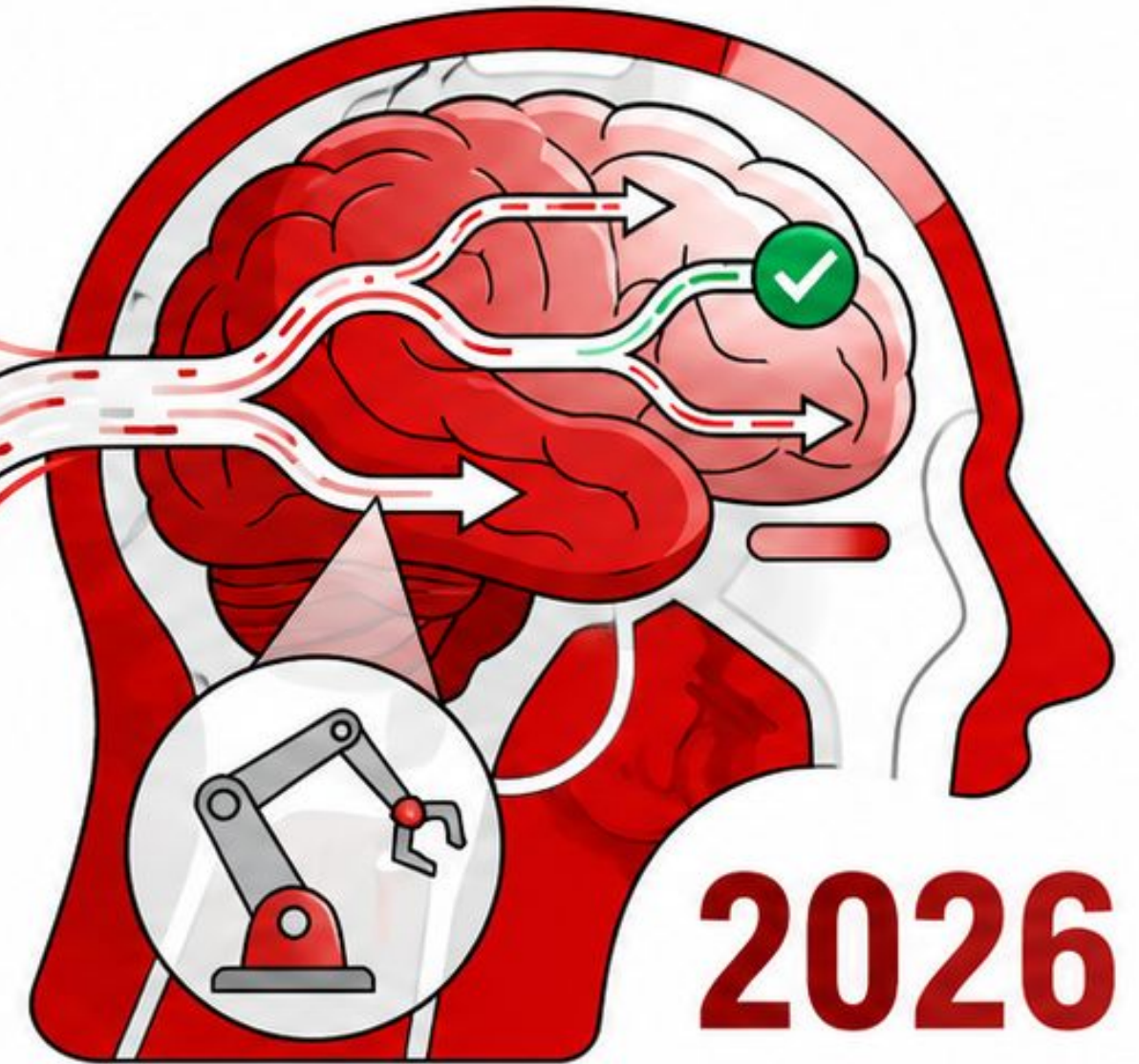
CONCLUSION: THE FUTURE IS REASONING



**STATIC RULES ARE FOR
STATIC SYSTEMS.**



**MODERN DELIVERY
REQUIRES A
“LIVING PLATFORM.”**



2026

**2026 IS THE YEAR WE
STOP “AUTOMATING” AND
START “REASONING.”**



ADAPTIVE INTELLIGENCE

Continuously learns, adapts,
and improves with context.



FASTER OUTCOMES

Smarter decisions and automation
lead to faster, more reliable delivery.



LOWER COST

Reduce waste, rework,
and operational overhead.



FUTURE-READY TEAMS

Empower people to focus on
innovation and impact.

Thank you! Questions?

